MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

THIRD SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG3DSCCFS200 –INTRODUCTION TO PROGRAMMING

Duration: 1.5 hrs MaximumMarks:50

Students should attempt at least one question from each course outcome to enhance their overall outcome attain ability.

Part A

Short Answer Type Questions

Answer **any 10** Questions

Each question carries **1** mark

1. Define the term “tokens” in the context of computer programming.[Understand][1]

2.Compare Algorithm and Flowchart [Analyse][2]

3.When overloading unary operators using Friend function,it requires_____ argument/s.. [Analyse][2]

4.Discuss about ADT. [Understand][1]

5.Identify :: the operator.[Analyse][2]

6.Differentiate while and do-while loop. [Analyse][2]

7. Explain what is abstract class?[Understand][1]

8.Explain what is a constructor?[Understand][1]

9.Explain what is inheritance? [Understand][1]

10.Explain what is the use of friend function?[Understand][1]

11.Select which of the following is a correct identifier in C++?

a)VAR_1234 b)$var_name c)7VARNAME d)7var_name [Evaluate][3]

12.Select which of the following symbol is used to declare the preprocessor directives in C++.

a)$ b) ^ c) # d)*[Evaluate][3]

[10x1=10]

Part B

Short Essay Type Questions

Answer **any 4** Questions

Each question carries **5** marks

13. Discuss the features of a good programming language .[Understand][1]

14. Compare and contrast call by value and call by reference using appropriate examples.[analyse][2]

15.Compare and contrast different types of inheritance in C++ [Analyse] [2]

16.Examine the use of Virtual Base class in C++ [Analyse] [2]

17.

a) What is friend Function in C++? [Understand][1]

b)Write a program to print Fibonacci series with friend function[Apply][4]

18.Distinguish between different types of constructors in C++.[Analyse][2]

[4x5=20]

Part C

Essay Type Questions

Answer **any 2** Questions

Each question carries **10** marks

19. Describe in detail about OOPs and POP concept with appropriate diagram and example. [Understand][1]

20.

a)What is inhertance ?Why we use Inheritance in C++?[Analyse][2]

b)Writea C++ program to read and print employee information using multiple inheritance.

[Apply][4]

21.

a)What is operator overloading in C++? What are the operators that cant be overloaded in C++? Mention Rules also.[Analyse][2]

b)Write a C++ program to overload binary operator '+' to add two complex numbers**.**

[Apply][4]

[2x10=20]